Capstone Project Report
Project Title: Procure-to-Pay (P2P) Cycle Automation Dashboard — A SAP MM Simulation for Zenith Industries Ltd.

COMPANY	Zenith Industries Ltd
INDUSTRIES	Quarterly supply order for production materials
SAP MODULE	SAP OTC / Order-to-Cash (O2C) Cycle
NAME	Sastika Deb
ROLL NO.	2307077
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Problem Statement

In most mid-sized manufacturing companies, the procurement process is handled manually through emails, spreadsheets, and disconnected approval chains. This creates five recurring problems:

● Lack of real-time visibility into purchase requisitions and open purchase orders.
● No systematic 3-way matching between Purchase Order, Goods Receipt, and Vendor Invoice, leading to duplicate or wrong payments.
● Delayed vendor payments damaging supplier relationships.
● Difficulty in tracking monthly procurement spend by vendor or material category.
● No centralized vendor or material master data.

The SAP MM module solves these problems in large enterprises, but understanding the complete P2P flow requires hands-on exposure. This project simulates the end-to-end P2P cycle of SAP MM as a web-based dashboard so that the process can be demonstrated, learned, and evaluated without needing a full SAP license.

Solution / Features

The project, **"P2P Dashboard — Zenith Industries Ltd."**, is a full-stack web application that digitally replicates the six stages of the SAP MM Procure-to-Pay cycle:

1. **Vendor Master Management** — Create and maintain vendor records (ID, payment terms, bank, tax).
2. **Material Master Management** — Maintain material codes, UoM, standard price, category.
3. **Purchase Requisition (PR)** — Requesters raise a PR; approvers approve/reject with remarks. Auto-numbered `10000001+`.
4. **Purchase Order (PO)** — Approved PRs are converted into POs and assigned to vendors. Auto-numbered `4500000001+`.
5. **Goods Receipt (GR)** — Warehouse posts GR against a PO; stock and GR document (`5000000001+`) are updated.
6. **Invoice Verification & Payment** — Accounts team enters vendor invoice; system performs **3-way match** (PO ↔ GR ↔ Invoice) before releasing payment. Invoice numbered `5100000001+`.

An executive **Dashboard** page shows KPIs, monthly spend trends, and top vendors. An **ALV-style Reports** page allows filtering and CSV export, imitating SAP's classic ALV report output.

Process Screenshots

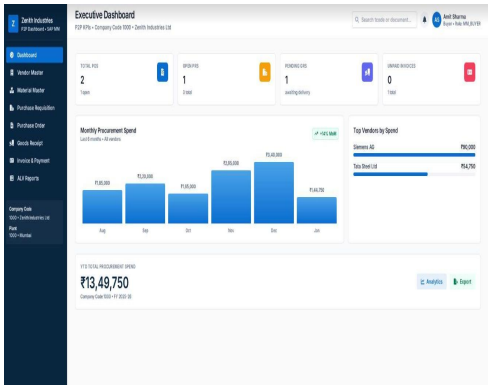

Fig 1:Executive Dashboard

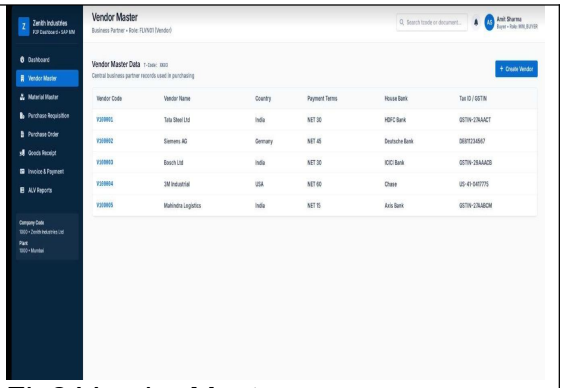

Fig2:Vendor Master

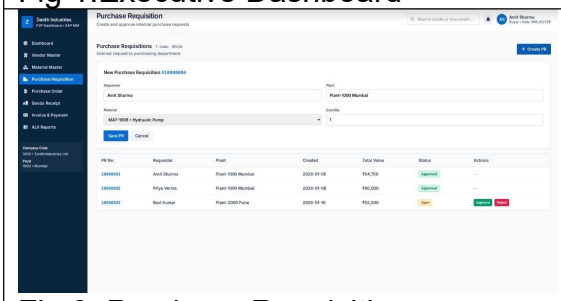

Fig 3: Purchase Requisition

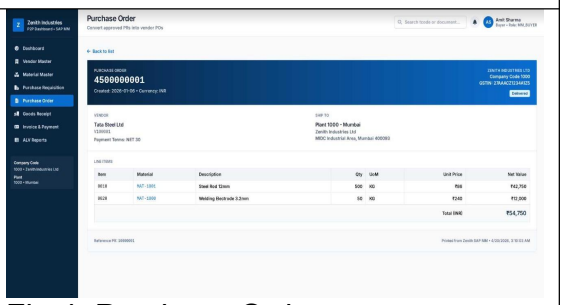

Fig 4: Purchase Order

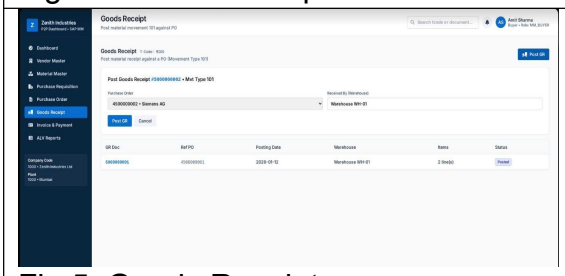

Fig 5: Goods Receipt

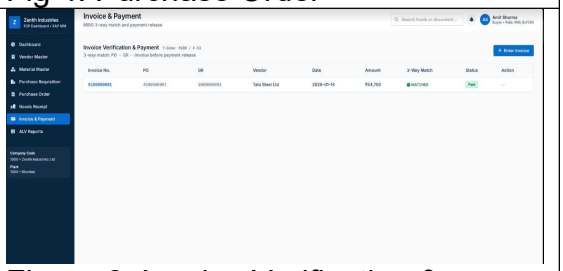

Figure 6: Invoice Verification & Payment

Tech Stack

Layer Technology
Frontend React 18, TailwindCSS, shadcn/ui, Recharts (charts), Lucide icons
Backend Python 3.11, FastAPI, Pydantic
Database MongoDB (NoSQL)
API Style REST with /api prefix
Auth JWT-based session (optional module)
Deployment Emergent Cloud Preview
Version Control Git & GitHub

Unique Points

Real SAP-style document numbering (PR 1000xxx, PO 4500xxx, GR 5000xxx, Invoice 5100xxx) matching SAP ECC/S4HANA conventions.
True 3-way matching logic implemented in the backend — payment is blocked if PO quantity ≠ GR quantity ≠ Invoice quantity, just like SAP.
Fiori-inspired UI (deep blue #0a6ed1, white tiles, left nav) giving an authentic SAP look & feel without needing SAP GUI.
Pre-seeded master data (5 vendors, 10 materials, sample documents) so the project is demo-ready on first launch.
ALV-style reports with CSV export, imitating SAP's signature list output.
Covers the full Procure-to-Pay cycle end-to-end in one application — most academic projects only cover one or two stages.

Future Improvements

1. **Integration with real SAP S/4HANA** via OData / BAPI calls so the dashboard can read/post real documents.
2. **Role-based access control** — separate logins for Requester, Approver, Buyer, Warehouse, Accounts.
3. **Automated approval workflow** using email notifications (SendGrid/Resend).
4. **AI-powered vendor recommendation** — suggest the best vendor for a material based on past price, delivery time, and quality score.
5. **OCR-based invoice capture** — upload a vendor invoice PDF and auto-extract invoice data using AI (Gemini Vision / OpenAI Vision).
6. **Mobile-responsive PWA** so approvers can approve PRs from their phone.
7. **Multi-currency & multi-company code** support for global rollouts.
8. **Integration with FI module** — auto-post accounting document on invoice verification.

Conclusion

This project translates the conceptual knowledge of SAP MM's Procure-to-Pay cycle into a functional, demonstrable software system. It gave hands-on exposure to enterprise process design, master data management, three-way matching, and financial posting — forming a strong foundation for a career as an SAP MM Functional or Techno-Functional consultant.